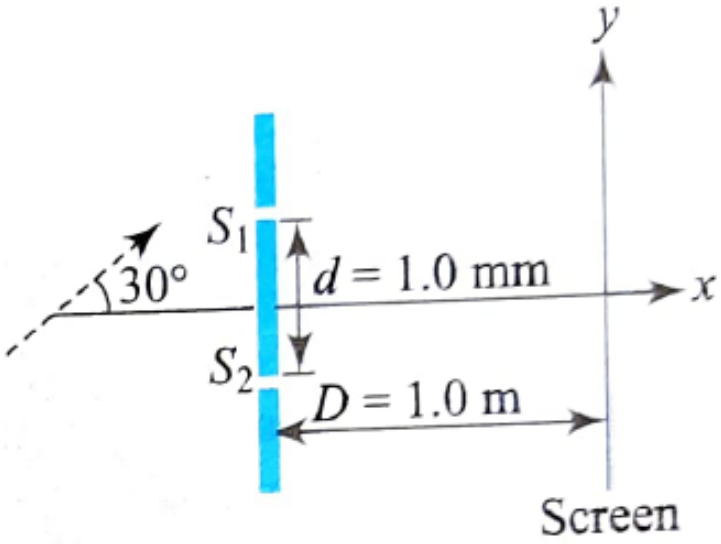
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A coherent parallel beam of wavelength $\lambda = 0.5$ mm falls on a Young's double-slit apparatus. The separation between the slits is 1.0 mm. The intensity of microwaves is measured on a screen placed parallel to the plane of the slits at a distance of 1.0 m from it as shown in figure.
If the incident beam makes an angle of 30° with the x-axis (as in the dotted arrow shown in the figure), find the y-coordinates of the first minima on either side of the central maximum.



(A) $\frac{3}{\sqrt{7}}$ and $\frac{1}{\sqrt{15}}$ m

(B) $\frac{3}{\sqrt{7}}$ and $\frac{1}{\sqrt{12}}$ m

(C) $\frac{3}{\sqrt{6}}$ and $\frac{1}{\sqrt{15}}$ m

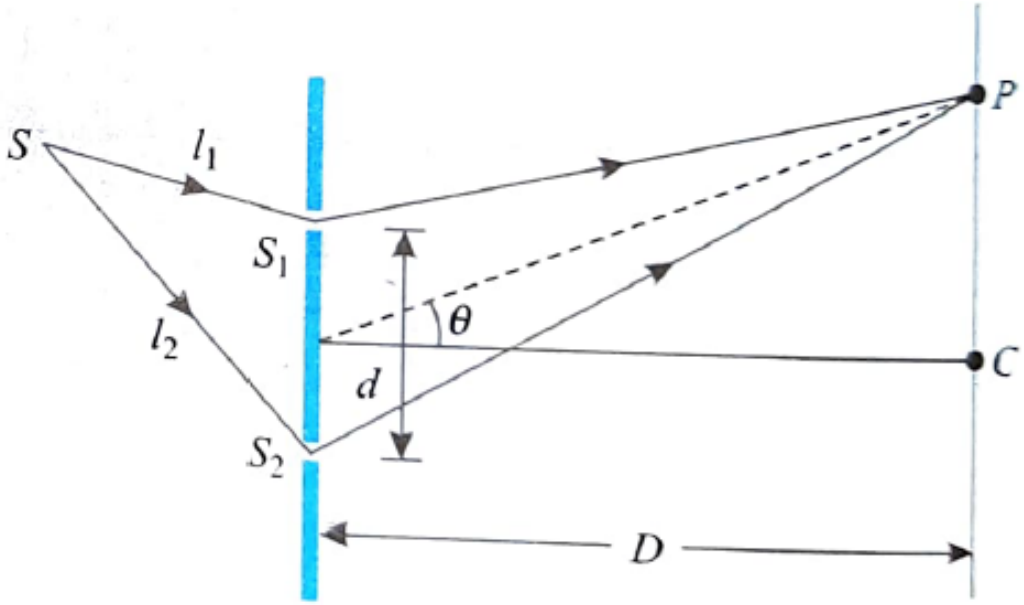
(D) $\frac{3}{\sqrt{7}}$ and $\frac{1}{\sqrt{16}}$ m

Q.2

In a Young's experiment the light source is at distance $l_1 = 20\mu$m and $l_2 = 40\mu$m from the slits. The light of wavelength $\lambda = 500$ nm is incident on slits separated at a distance 10μm. A screen is placed at a distance $D = 2$ m away from the slits as shown in figure.
(a) Find the values of θ relative to the central line where maxima appear on the screen.

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(b) How many maxima will appear on the screen?



(A) $\sin^{-1}\left[2\left(\frac{n}{40}-1\right)\right]$, 24

(B) $\sin^{-1}\left[2\left(\frac{n}{80}-1\right)\right]$, 57

(C) $\sin^{-1}\left[2\left(\frac{n}{60}-1\right)\right]$, 37

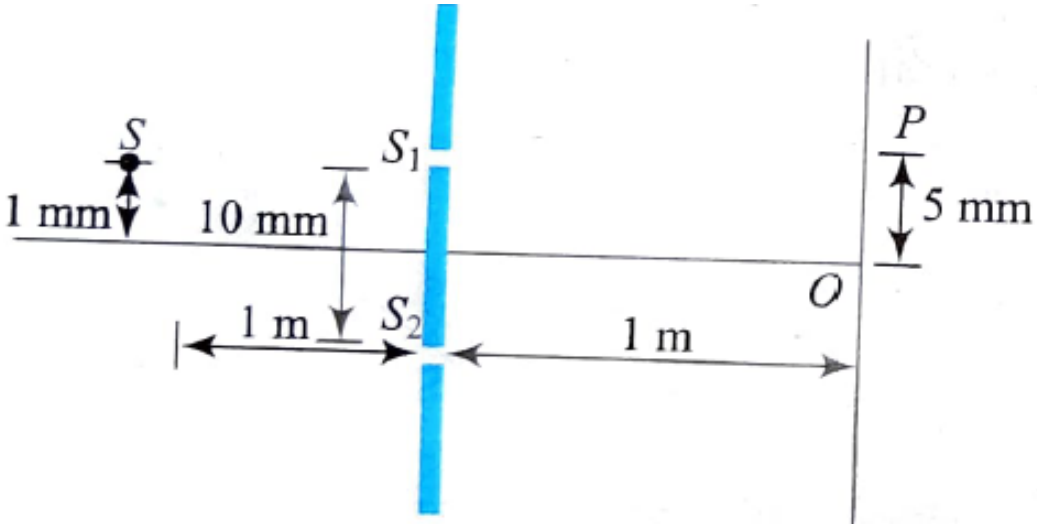
(D) $\sin^{-1}\left[2\left(\frac{n}{40}-1\right)\right]$, 40

Q.3

In Young's double-slit experiment, the point source is placed slightly off the central axis as shown in figure.

(a) Find the nature and order of the interference at point P.

(b) Find the nature and order of the interference at point O.



(A) 20

(B) 54

(C) 37

(D) 40

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Q.4

Acoherent parallelbeam ofmicrowaves of wavelength $0.5\ \text{mm}$ falls normally on Young's double slit apparatus. The separation between the slits is $1.0\ \text{mm}$ and the screen is placed at a distance of $1.0\ \text{m}$ from the slits. The number of minima in the interference pattern observed on the screen is

(A) 3

(B) 4

(C) 5

(D) much greater than 5

Q.5

A beam of light consisting of two wavelengths $\lambda_1 = 750\ \text{nm}$ and $\lambda_2 = 450\ \text{nm}$ is used in Young's double slit experiment. The separation between the slits is $2\ \text{mm}$ and the distance of the screen from the plane of the slits is $100\ \text{cm}$. What is the minimum distance between two successive regions of maximum brightness of two wavelengths?

(A) $1.125\ \text{mm}$

(B) $2.250\ \text{mm}$

(C) $3.525\ \text{mm}$

(D) $4.750\ \text{mm}$

Q.6

In Young's double slit experiment, the intensity at a point P on the screen is $3/4$ of the maximum intensity in the interference pattern. If d is the separation between the slits and λ is the wavelength of light, the angular separation between point P and the centre of the screen is

(A) $\sin^{-1}\left(\frac{\lambda}{d}\right)$

(B) $\cos^{-1}\left(\frac{\lambda}{3d}\right)$

(C) $\tan^{-1}\left(\frac{\lambda}{6d}\right)$

(D) $\sin^{-1}\left(\frac{2\lambda}{3d}\right)$

Q.7

In Young's double slit experiment using monochromatic light, the fringe pattern shifts by a certain distance on the screen when a transparent sheet of thickness t and refractive index μ is introduced in the path of one of the interfering waves. The sheet is then removed and the distance between the screen and the slits is doubled. It is found that the distance between successive maxima (or minima) now is the same as the observed fringe shift when the sheet was introduced. The wavelength of light used is

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(A) $\lambda = (\mu - 1)t$

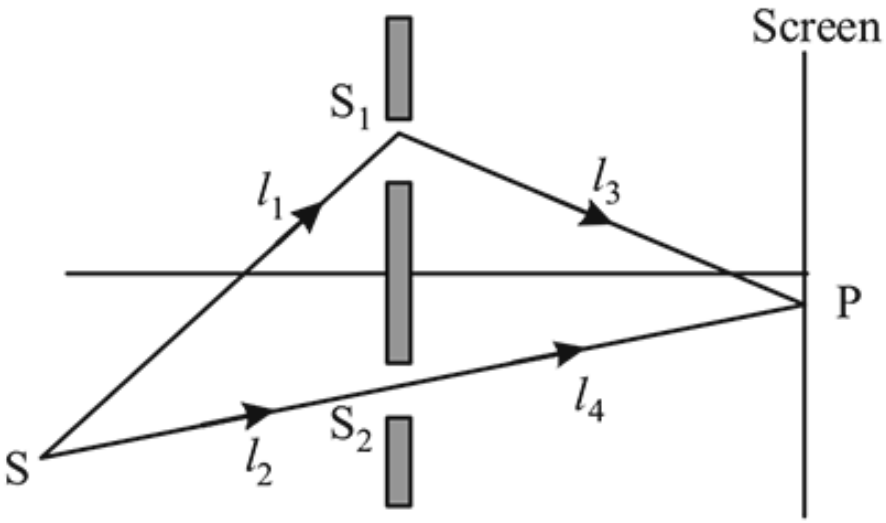
(B) $\lambda = \frac{1}{2}(\mu - 1)t$

(C) $\lambda = (\mu + 1)t$

(D) $\lambda = \frac{1}{2}(\mu + 1)t$

Q.8

The YDSE apparatus is as shown in Fig. The condition for point P to be a dark fringe is



(A) $(l_1 - l_3) + (l_2 - l_4) = \mathrm{n}\lambda$

(B) $(l_1 - l_2) + (l_3 - l_4) = \frac{(2\mathrm{n}-1)}{2}\lambda$

(C) $(l_1 - l_3) + (l_2 - l_4) = \frac{(2\mathrm{n}-1)\lambda}{2}$

(D) $(l_1 - l_2) + (l_4 - l_3) = \frac{(2n-1)\lambda}{2}$

Q.9

In Young's double slit experiment intensity at a point is how much times of the maximum intensity if angular position of this point is $\sin^{-1}(\lambda/6\ \mathrm{d})$?

(A) $1/2$

(B) $2/5$

(C) $3/4$

(D) $3/5$

Q.10

A monochromatic light is used in Young's double slit experiment when one of the slits is covered by a transparent sheet of thickness $1.8\ \mathrm{mm}$, made of material of refractive index μ_1 number of fringes which

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shift is 18 , when another sheet of thickness $3.6\ \mathrm{mm}$, made of material of refractive index μ_2 is used, number of fringes which shift is 9. Relation between μ_1 and μ_2 is given by

(A) $4\mu_2 - \mu_1 = 3$

(B) $4\mu_1 - \mu_2 = 3$

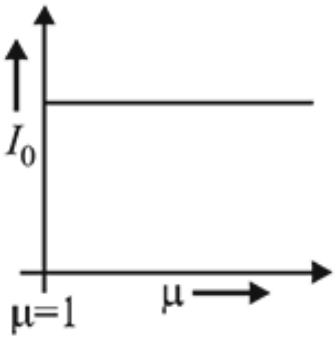
(C) $3\mu_2 - \mu_1 = 4$

(D) $2\mu_1 - \mu_2 = 4$

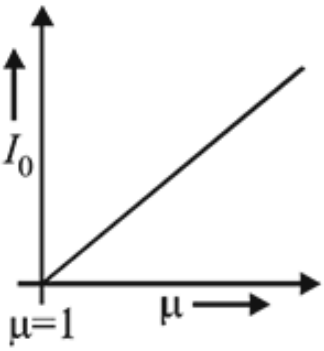
Q.11

In a $YDSE$ experiment if a slab whose refraction index can be varied is placed in front of one of the slits then the variation of resultant intensity at midpoint of screen with ' μ ' will be best represented by $(\mu \geq 1)$. [Assume slits of equal width and there is no absorption by slab; mid point of screen is the poin where waves interfere with zero phase difference in absence of slab]

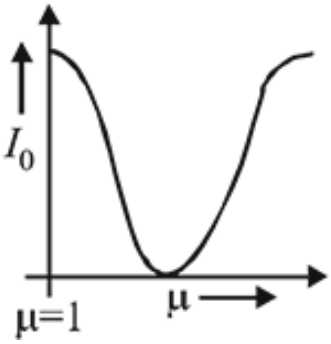
(A)



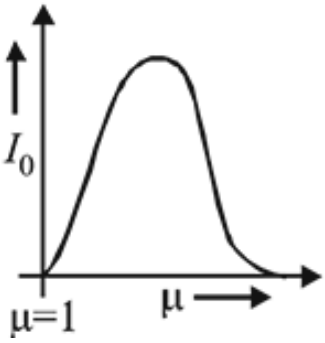
(B)



(C)



(D)



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Q.12

In young's double - slit experiment, the separation between the slits is d, distance between the slit and screen is $D(D >> d)$. In the interference pattern, there is a maxima exactly in front of each slit. Then the possible wavelength(s) used in the experiment are

(A) $\mathrm{d}^2/\mathrm{D}, \mathrm{d}^2/2\mathrm{D}, \mathrm{d}^2/3\mathrm{D}$

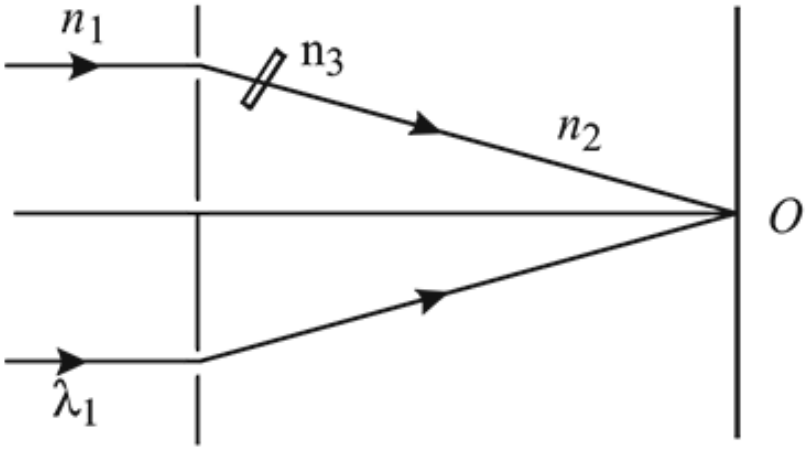
(B) $\mathrm{d}^2/\mathrm{D}, \mathrm{d}^2/3\mathrm{D}, \mathrm{d}^2/5\mathrm{D}$

(C) $\mathrm{d}^2/2\mathrm{D}, \mathrm{d}^2/4\mathrm{D}, \mathrm{d}^2/6\mathrm{D}$

(D) None of these

Q.13

In the figure shown in a $YDSE$, a parallel beam of light is incident on the slits from a medium of refractive index n_1. The wavelength of light in this medium is λ_1. Atransparent slab of thickness and refractive index is put infront of one slit. The medium between the screen and the plane of the slits is n_2. The phase difference between the light waves reaching point O (symmetrical, relative to the slit) is



(A) $\frac{2\pi}{n_1\lambda_1}(n_3 - n_2)\,t$

(B) $\frac{2\pi}{\lambda_1}(n_3 - n_2)\,t$

(C) $\frac{2\pi n_1}{n_2\lambda_1}\left(\frac{n_3}{n_2} - 1\right)t$

(D) $\frac{2\pi n_1}{\lambda_1}(n_3 - n_2)\,t$

Q.14

In a Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of same thickness, the shift of fringes is $(3/2)x$. The refractive index of second plate is

(A) 1.75

(B) 1.50

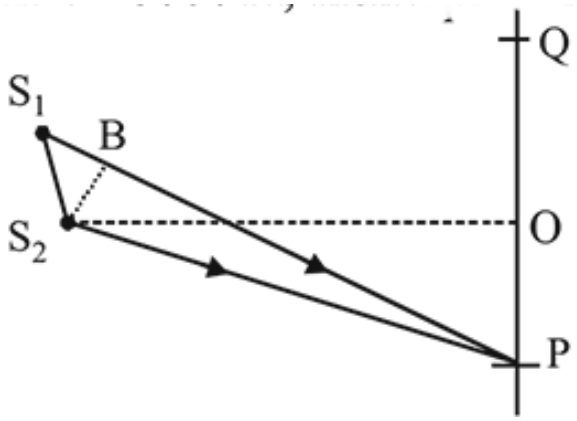
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(C) 1.25

(D) 1.00

Q.15

In figure, Young's double slit experiment Q is the position of the first bright fringe on the right side of O. P is the 11th bright fringe on the other side, as measured from Q. If $\lambda = 6000\mathring{A}$, then $\mathrm{S_1\ B}$ wil be



(A) $6 \times 10^{-6}\ \mathrm{m}$

(B) $6.6 \times 10^{-6}\ \mathrm{m}$

(C) $3.318 \times 10^{-7}\ \mathrm{m}$

(D) $3.144 \times 10^{-7}\ \mathrm{m}$

Q.16

The maximum number of possible interference maxima for slit separation equal to 1.8λ, where λ is the wavelength of light used, in a Young's double slit experiment is

(A) zero

(B) 3

(C) infinite

(D) 5

Q.17

In a Young's double slit experiment, if the incident light consists of two wavelengths λ_1 and λ_2, the slit separation is d, and the distance between the slit and the screen is D, the maxima due to the two wavelengths will coincide at a distance from the central maxima, given by:

(A) $\frac{\lambda_1\lambda_2}{2\mathrm{Dd}}$

(B) $(\lambda_1 - \lambda_2) \cdot \frac{2\,\mathrm{d}}{\mathrm{D}}$

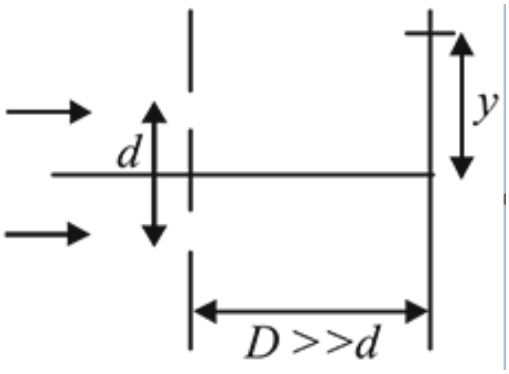
(C) LCM of $\lambda_1 \cdot \frac{D}{d}$ and $\lambda_2 \cdot \frac{D}{d}$

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(D) HCF of $\frac{\lambda_1 D}{d}$ and $\frac{\lambda_2 D}{d}$

Q.18

Consider the YDSE arrangement shown in figure. If $d = 10\lambda$ then position of 8th maxima is



(A) $y = \frac{D}{10}$

(B) $y = \frac{D}{3}$

(C) $y = \frac{4}{5}D$

(D) $y = \frac{4D}{3}$

Q.19

In a double slit experiment interference is obtained from electron wave produced in an electron gun supplied with voltage V. If λ is wavelength of the beam, D is the distance of screen d is the spacing between coherent sources, h is plank's constant, e is charge on electron and m is mass of electron, then fringe width is given as :-

(A) $\frac{hD}{\sqrt{2\ meVd^2}}$

(B) $\frac{2\ hD}{\sqrt{meVd^2}}$

(C) $\frac{hd}{\sqrt{2\ meVD^2}}$

(D) $\frac{2\ hd}{\sqrt{meVD^2}}$

Q.20

Visible light of wavelength 6000×10^{-8}cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction minimum is at 60° from the central maximum. If the first minimum is produced at θ_1, then θ_1 is close to

(A) 20°

(B) 30°

(C) 25°

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(D) 45°

Answers & Solutions

Q.1 Answer:

$\frac{3}{\sqrt{7}}$ and $\frac{1}{\sqrt{15}}$ m

Solution:

Path difference before slits $= d\sin\phi$

Path difference after slits $= d\sin\theta$ **As path of rays before slits is longer at** S_1 **and** $S_2P > S_1P$ **after slits, so net path difference for first minima is**

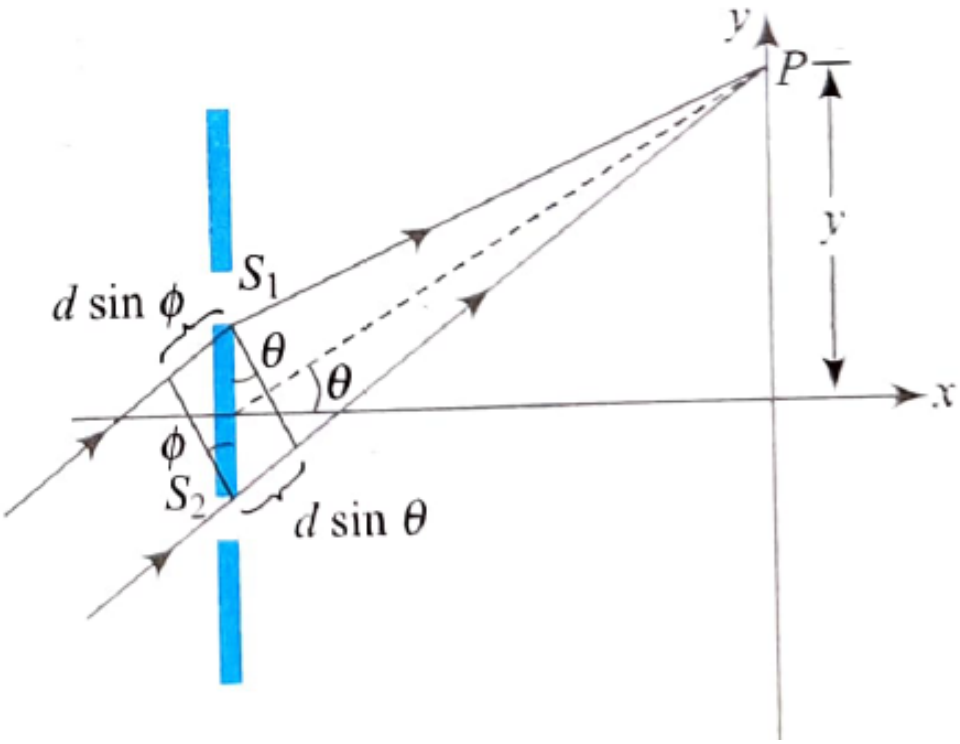
$$\Delta x = d\sin\theta - d\sin\phi = \pm\frac{\lambda}{2}$$

$$\Rightarrow \quad \sin\theta = \sin\phi \pm \frac{\lambda}{2d} = \sin 30^\circ \pm \frac{0.5}{2\times 1} = \frac{3}{4} \text{ or } \frac{1}{4}$$

$$\Rightarrow \quad \tan\theta = \frac{3}{\sqrt{7}} \text{ and } \frac{1}{\sqrt{15}}$$

So, the position of first minima on either side of the central maxima is

$$y = D\tan\theta = \frac{3}{\sqrt{7}} \text{ and } \frac{1}{\sqrt{15}} \text{ m}$$



Q.2 Answer:

$\sin^{-1}\left[2\left(\frac{n}{40}-1\right)\right]$, 40

Solution:

(a) The optical path difference between the beams artiving at P, $\Delta x = (l_2 - l_1) + d\sin\theta$

The condition for maximum intensity is, $\Delta r = n\lambda$

where, $n = 0, \pm 1, \pm 2, \ldots$

Thus, $\sin\theta = \frac{1}{d}\left[\Delta x - (l_2 - l_1)\right] = \frac{1}{d}\left[n\lambda - (l_2 - l_1)\right]$

$= \frac{1}{10\times 10^{-6}}\left[n\times 500\times 10^{-9} - 20\times 10^{-6}\right] = 2\left[\frac{n}{40} - 1\right]$

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hence, $\theta = \sin^{-1}\left[2\left(\frac{n}{40} - 1\right)\right]$

(b) $\sin\theta \,| \leq 1$

$-1 \leq 2\left[\frac{n}{40} - 1\right] \leq 1$

of $-20 \leq (n - 40) \leq 20$

$20 \leq n \leq 60$

Hence, number of maxima $= 60 - 20 = 40$

Q.3 Answer:

20

Solution:

(a) As discussed in previous case, here $b = 1\ \text{mm}, d = 10\ \text{mm}, a = 1\ \text{m}$, **and** $D = 1\ \text{m}$.

Using Eq. (i) for path difference between the waves reaching at point P,

$\Delta x = \frac{1.10}{10^3} + \frac{5\times10}{2\times10^3} = 35 \times 10^{-3}\ \text{mm}$

If it is maximum at point P, **then** $\Delta x/\lambda$ **will be an integer,**

i.e., $\frac{35\times10^{-3}}{500\times10^{-9}\times10^3} = 70$

A maximum of the order 70 occurs at P.

At $O, \Delta x_0' = \frac{1\times10}{10^3} = 10 \times 10^{-3}\ \text{mm}$

$\Rightarrow \quad \frac{\Delta x}{\lambda} = 20$

Hence, maximum of the order 20 occurs at O.

Q.4 Answer:

4

Solution:

Refer to Fig. When the incident beam falls normally on the slits S_1 **and** S_2, **the path difference at the central point** P_0 **of the screen is zero. Hence we have the central maximum at** P_0.

Let the minima appear along directions θ **with respect to the incident direction. Coherent waves from** S_1 **and** S_2 **along this direction are brought to a focus at** P. **It is clear that the path difference between the waves from** S_1 **and** S_2 **on reaching** P **is**

$\Delta = d\sin\theta$

The interference minima will appear on the screen if

$\Delta = \left(n + \frac{1}{2}\right)\lambda$

or $d\sin\theta = \left(n + \frac{1}{2}\right)\lambda; n = 0, \pm1, \pm2, \ldots$.

Thus the directions of minima are given by

$\sin\theta = \left(n + \frac{1}{2}\right) \times \left(\frac{\lambda}{d}\right)$

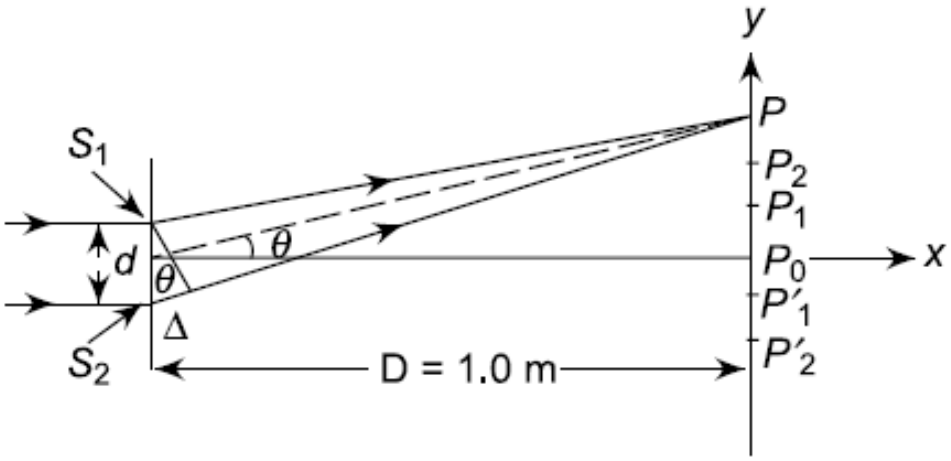
Given $d = 1.0\ \text{mm}$ **and** $\lambda = 0.5\ \text{mm}$. **Therefore**

$\sin\theta = \left(n + \frac{1}{2}\right) \times \left(\frac{0.5}{1.0}\right) = \frac{1}{2}\left(n + \frac{1}{2}\right)$

The allowed values of m **are those integers for which** $\sin\theta$ **is not more than** $+1$ **or less than**

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-1. **These values are** $n = 1, 0, -1$ **and** -2. **Hence four minima will be observed. The correct choice is (b).**



Q.5 Answer:

1.125 mm

Solution:

$y_1 = \frac{n\lambda_1 \Delta}{d}$ **(1)**

$y_2 = \frac{m\lambda_2 \Delta}{d}$ **(2)**

The waves will produce maximum brightness at the screen where $y_1 = y_2$, **i.e.,**

$$\frac{n\lambda_1 D}{d} = \frac{m\lambda_2 D}{d}$$

$$\Rightarrow \quad \frac{n}{m} = \frac{\lambda_2}{\lambda_1} = \frac{450 \text{ nm}}{750 \text{ nm}} = \frac{3}{5}$$

The minimum integral values of n **and** m **which satisfy this condition are** $n_1 = 3$ **and** $m_1 = 5$ **and the next values are** $n_2 = 6$ **and** $m_2 = 10$. **So, we get the first region of maximum brightness when the 3rd bright fringe of** λ_1 **falls on the 5 th bright fringe of** λ_2.

$$(\Delta y)_1 = \frac{3\lambda_1 D}{d} = \frac{5\lambda_2 D}{d}$$

$$= \frac{3 \times \left(750 \times 10^{-9}\right) \times 1.0}{2 \times 10^{-3}} = 1.125 \text{ mm}$$

We get the next region of maximum brightness when the 6 th bright fringe of λ_1 **falls on the 10 th bright fringe of** λ_2

$$(\Delta y)_2 = \frac{6\lambda_1 D}{d} = \frac{10\lambda_2 D}{d}$$

$$= \frac{6 \times \left(750 \times 10^{-9}\right) \times 1.0}{2 \times 10^{-3}} = 2.25 \text{ mm}$$

$$(\Delta y)_{\min} = (\Delta y)_2 - (\Delta y)_1$$

$$= 2.25 - 1.125$$

$= 1.125$ mm, **which is choice (a).**

Q.6 Answer:

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$\tan^{-1}\left(\frac{\lambda}{6d}\right)$

Solution:

In Young's double slit experiment,

$I = 4I_0 \cos^2(\phi/2)$

where ϕ is the phase difference between the interfering beams.

$I_{\max} = 4I_0$

Given $I = \frac{3}{4} I_{\max}$

$$\text{or} \quad 4I_0 \cos^2\left(\frac{\phi}{2}\right) = \frac{3}{4} I_{\max}$$

$$\Rightarrow \quad \cos^2\left(\frac{\phi}{2}\right) = \frac{3}{4}$$

$$\Rightarrow \cos\left(\frac{\phi}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \quad \frac{\phi}{2} = \frac{\pi}{6} \quad \text{or } \phi = \frac{\pi}{3}$$

Now refer to Fig.

If θ is the angular separation between P and P_0,

$\tan\theta = \frac{y}{D}$

If β is the fringe width between two consecutive maxima, than $\beta = 2\pi$. Hence for point P

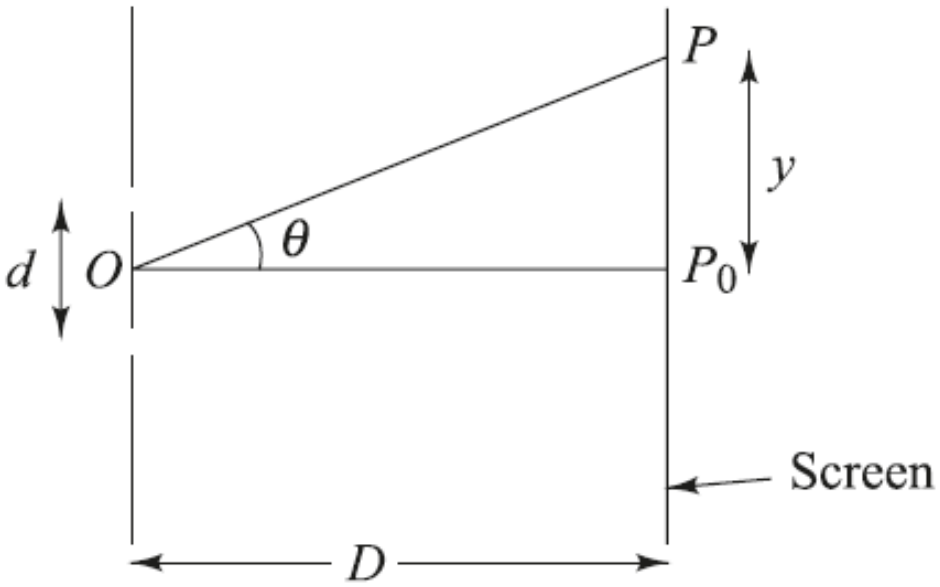
$$\frac{y}{\beta} = \frac{\pi/3}{2\pi} = \frac{1}{6}$$

$$\text{or } \frac{yd}{D\lambda} = \frac{1}{6}$$

or $\tan\theta \times \frac{d}{\lambda} = \frac{1}{6}$

or $\tan\theta = \frac{\lambda}{6d} \Rightarrow \theta = \tan^{-1}\left(\frac{\lambda}{6d}\right)$

So the correct choice is (c).



Q.7 Answer:

$\lambda = \frac{1}{2}(\mu - 1)t$

Solution:

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When a sheet of thickness t and refractive index μ is introduced in one of the interfering waves, the distance y_0 through which the fringes shift is given by

$y_0 = (\mu - 1)t\frac{D}{d}$

The fringe width β, i.e., the distance between successive maxima (or minima) is given by

$\beta = \frac{\lambda D}{d}$

When the distance D between the slits and the screen is doubled, the new fringe width becomes

$\beta' = 2\frac{\lambda D}{d}$

It is given that $y_0 = \beta'$. Equating Eqs. (1) and (2) we get

$$2\frac{\lambda D}{d} = (\mu - 1)t\frac{D}{d}$$
$$\lambda = \frac{1}{2}(\mu - 1)t$$

which is choice (b)

Q.8 Answer:

$(l_1 - l_2) + (l_3 - l_4) = \frac{(2n-1)}{2}\lambda$

Solution:

For dark frlindge, the path difference should be old multiple of $\mathrm{N}/2$.

So, $(l_1 + l_3) - (l_2 + l_4) = (2n - 1)\frac{\lambda}{2}$

The correct answer is (c) $(l_1 + l_3) - (l_2 + lu) = (2n - 1)\frac{\lambda}{2}$

Q.9 Answer:

$3/4$

Solution:

Angular position $\theta = \sin^{-1}\left(\frac{\lambda}{6d}\right)$

Path difference $= \Delta x = d\sin\theta$

$\Delta x = \frac{\lambda}{6}$

W.k.T,

$$\Delta x = \frac{\lambda}{2\pi}\Delta\phi$$
$$\frac{\lambda}{6} = \frac{\lambda}{2\pi}\Delta\phi$$
$$\Delta\phi = \frac{\pi}{3}$$

Intensity at a point $I = I_{\max}\cos^2\left(\frac{\Delta\phi}{2}\right)$

$$= I_{\max}\cos^2\left(\frac{\pi}{6}\right)$$
$$= I_{\max}\left(\frac{3}{4}\right)$$

Q.10 Answer:

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$4\mu_2 - \mu_1 = 3$

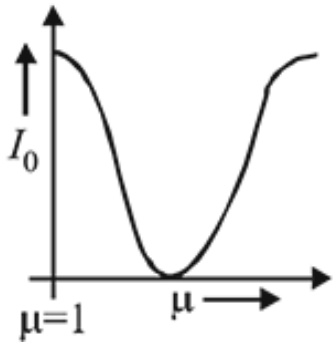
Solution:

$(\mu - 1)t = n\beta$

$\frac{(\mu_1 - 1) \times 1.8 \times 10^{-5}}{(\mu_2 - 1) \times 3.6 \times 10^{5}} = \frac{18\beta}{9\beta}$

$(\mu_1 - 1) = 4\,(\mu_2 - 1) \quad 4\mu_2 - \mu_1 = 3$

Q.11 Answer:



Solution:

In absence of film or for $m = 0$ intensity is maximum at screen. As the value of m is increased, intensity shall decrease and then increase alternately. Hence the correct variation is (c).

Q.12 Answer:

$\mathrm{d}^2/2\mathrm{D}, \mathrm{d}^2/4\mathrm{D}, \mathrm{d}^2/6\mathrm{D}$

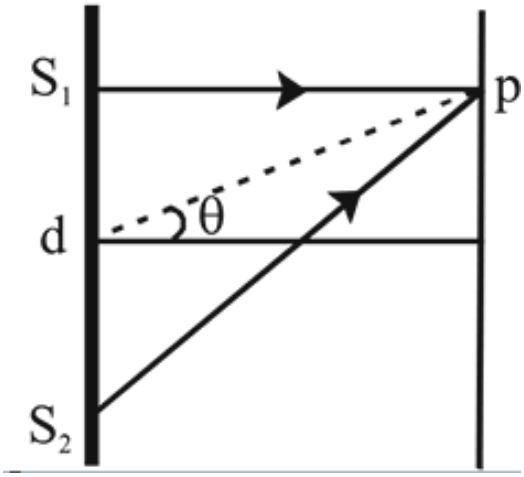
Solution:

$S_2P - S_1P = \frac{dy}{D} = \frac{d \times (d/2)}{D} = \frac{d^2}{2D}$

$\frac{d^2}{2D} = n\lambda$ S_1

$\lambda = \frac{d^2}{2nD}, n = 1, 2, \ldots\ldots$ d

$\lambda = \frac{d^2}{2D}, \frac{d^2}{4D}, \frac{d^2}{6D}$ S_2



Q.13 Answer:

$\frac{2\pi}{n_1\lambda_1}(n_3 - n_2)\,t$

Solution:

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Optical path difference between the waves $= (n_3 - n_2)\, t$

- Phase difference

$= 2\pi \frac{(n_3 - n_2)t}{\lambda_{(\text{Vacuum})}} = 2\pi \frac{(n_3 - n_2)t}{n_1 \lambda_1}$

Q.14 Answer:

1.75

Solution:

Using relation, $y_0 = \frac{D}{d}(\mu - 1)t$

We have, $\frac{z}{\frac{3}{2}z} = \frac{(1.5-1)}{(\mu-1)} \Rightarrow \frac{2}{3} = \frac{1}{2(\mu-1)}$

$\frac{2}{\mu-1} = \frac{4}{3} \Rightarrow \mu - 1 = \frac{3}{4}$

$\mu = \frac{7}{4} = 1.75$

Q.15 Answer:

$6 \times 10^{-6}\ \text{m}$

Solution:

Path difference between first and 11th **bright fringe** $= \text{S}_1\ \text{B}(10$ **bright fringes** $) = 10\lambda$

$= 10 \times \left(6000 \times 10^{-7}\right)\ \text{m} = 6 \times 10^{-6}\ \text{m}$

Q.16 Answer:

3

Solution:

As $\sin\theta = \frac{n\lambda}{d}$ **and** $\sin\theta$ **cannot be** $\not{1}$

$\therefore 1 = \frac{n\lambda}{1.8\lambda}$ **or** $n = 1.8$

Hence maximum number of possible interference maximas, $0, \pm 1$ **i.e. 3**

Q.17 Answer:

LCM of $\lambda_1 \cdot \frac{D}{d}$ and $\lambda_2 \cdot \frac{D}{d}$

Solution:

Correct option is (b) LCM of $\frac{\lambda_1 \text{D}}{\text{d}}$ **and** $\frac{\lambda_2 \text{D}}{\text{d}}$

Let n_1 **bright order of wavelenth** λ_1 **coincides with** n_2 **bright order of wavelenth** λ_2

Its distance from central maximum $\text{y} = \frac{\text{n}_1 \lambda_1 \text{D}}{\text{d}} = \frac{\text{n}_2 \lambda_2 \text{D}}{\text{d}}$ **which gives LCM of** $\frac{\lambda_1 \text{D}}{\text{d}}$ **and** $\frac{\lambda_2 \text{D}}{\text{d}}$

Q.18 Answer:

$y = \frac{4D}{3}$

Solution:

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For maxima $d\sin\theta = n\lambda$

$\sin\theta = \frac{n\lambda}{d} = \frac{8\lambda}{10\lambda} \Rightarrow \sin\theta = \frac{4}{5} \Rightarrow \tan\theta = \frac{4}{3}$

Also $\tan\theta = \frac{y}{D} \quad \therefore \quad y = \frac{4D}{3}$

Q.19 Answer:

$\frac{hD}{\sqrt{2\ \text{meVd}^2}}$

Solution:

$\beta = \frac{\lambda D}{d}, \quad \lambda = \frac{h}{P} = \frac{h}{\sqrt{2\ \text{m}(KE)}} = \frac{h}{\sqrt{2\ \text{meV}}}$

$\therefore \beta = \frac{\lambda D}{\sqrt{2\ \text{meV}}}\frac{1}{d} = \frac{hD}{\sqrt{2\ \text{meVd}^2}}$

Q.20 Answer:

25°

Solution:

For 2nd **minima,**

$d\sin\theta = 2\lambda$

$\sin\theta = \frac{\sqrt{3}}{2}$ **, (given)**

$\Rightarrow \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$ **...(1)**

So, for 1st **minima**

$d\sin\theta = \lambda$

$\Rightarrow \sin\theta = \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$

$\Rightarrow \theta = 25.65^o$

$\Rightarrow \theta \approx 25^o$

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